The Theory of Computing

Turing Machines

ON COMPUTABLE NUMBERS, WITH AN APPLICATION TO THE ENTSCHEIDUNGSPROBLEM

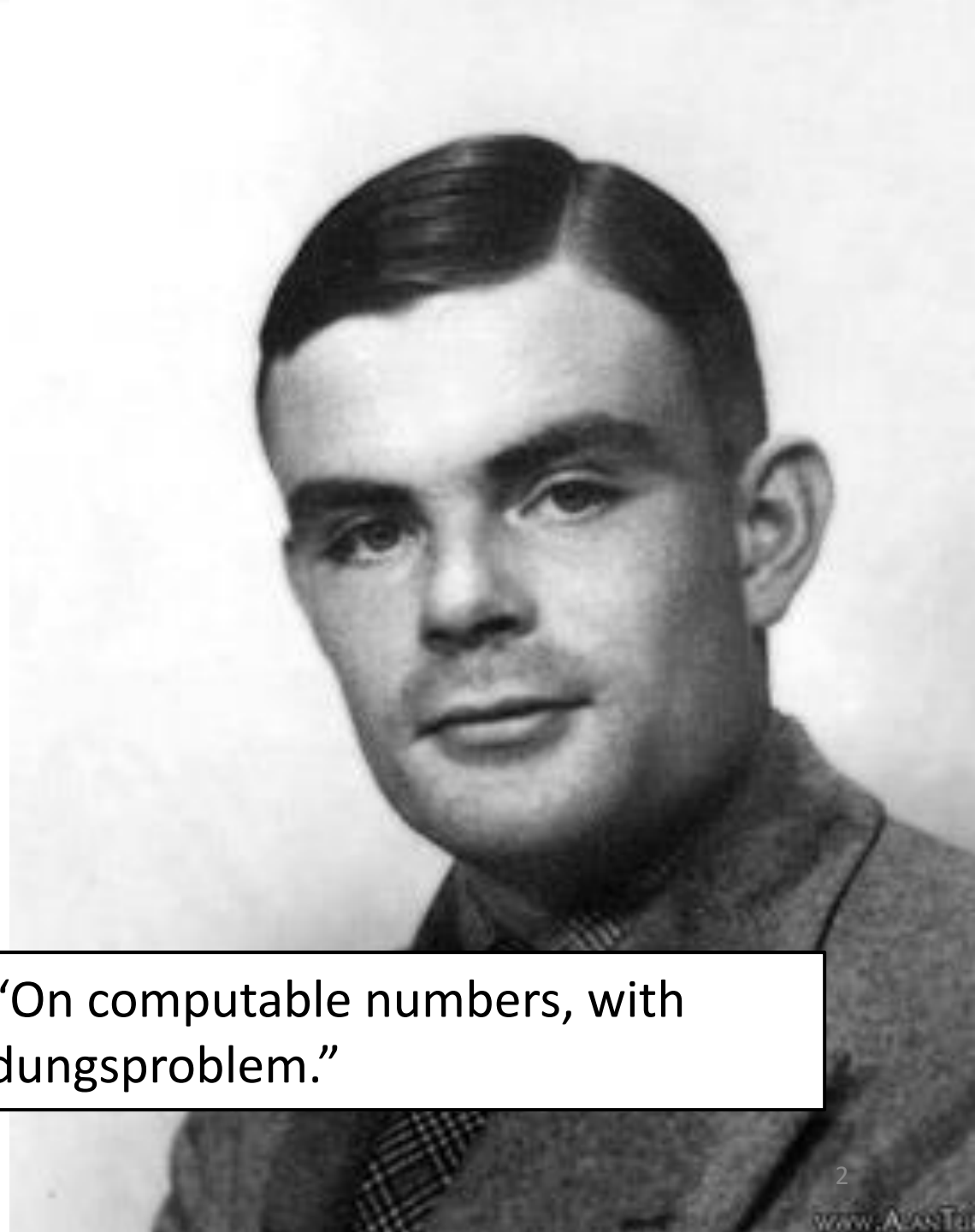
By A. M. TURING.

[Received 28 May, 1936.—Read 12 November, 1936.]

The "computable" numbers may be described briefly as the real numbers whose expressions as a decimal are calculable by finite means. Although the subject of this paper is ostensibly the computable *numbers*, it is almost equally easy to define and investigate computable functions of an integral variable or a real or computable variable, computable predicates, and so forth. The fundamental problems involved are, however, the same in each case, and I have chosen the computable numbers

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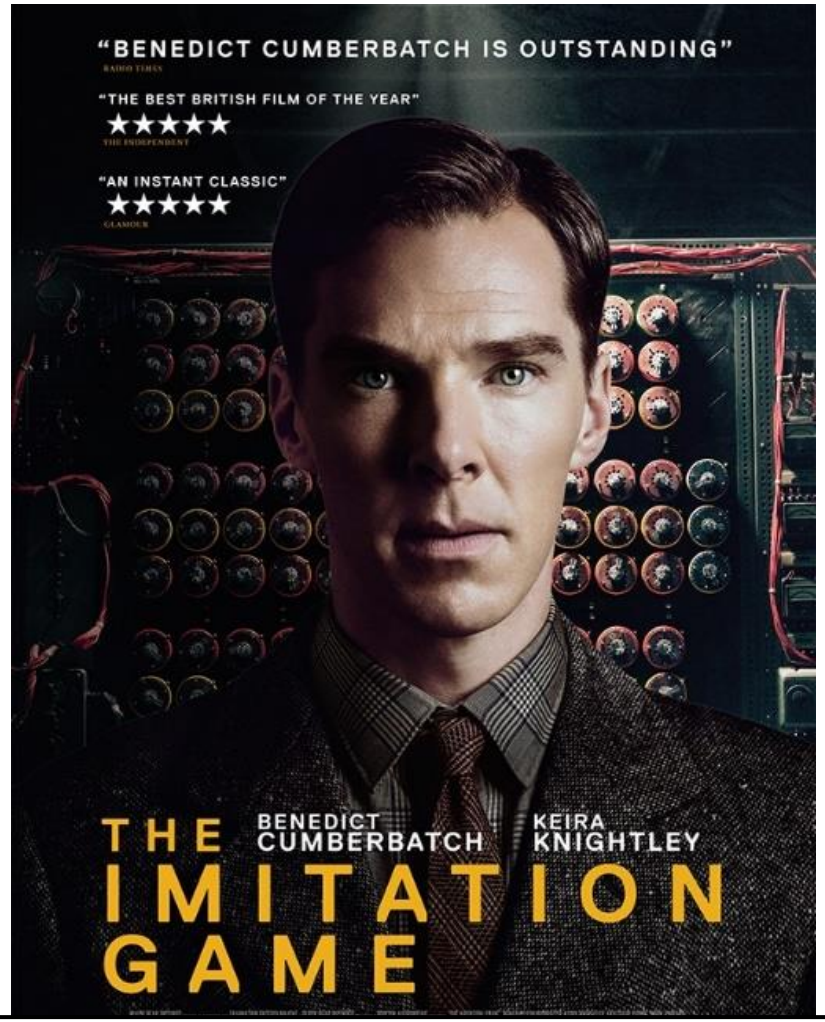
In 1936, Alan Turing published the paper "On computable numbers, with application to the Entsheidungsproblem."

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This paper is briefly mentioned at the beginning of "The Imitation Game", a movie about Alan Turing's life and contribution to codebreaking in World War II.

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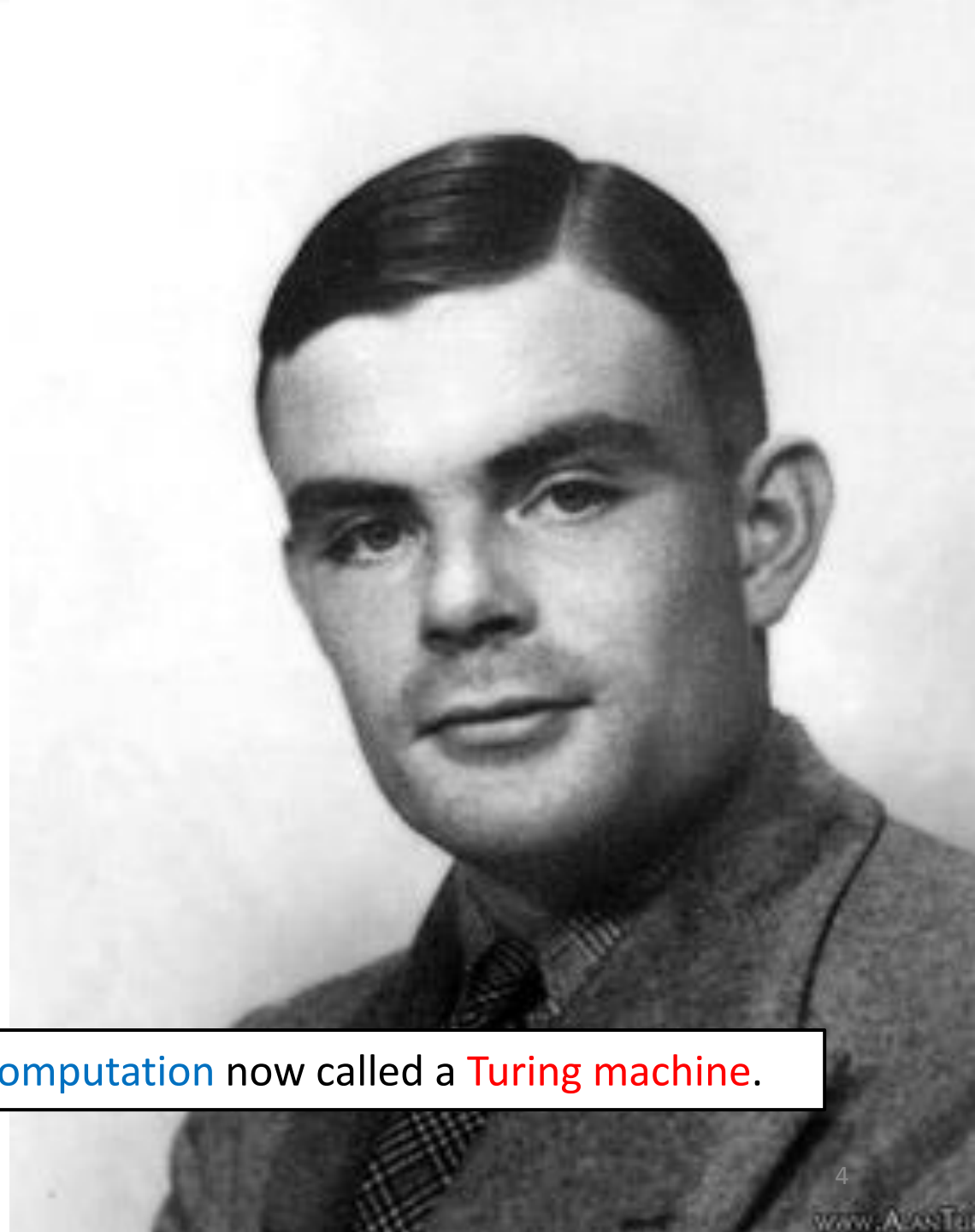
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In this paper, Alan Turing introduced a new model of computation now called a Turing machine.

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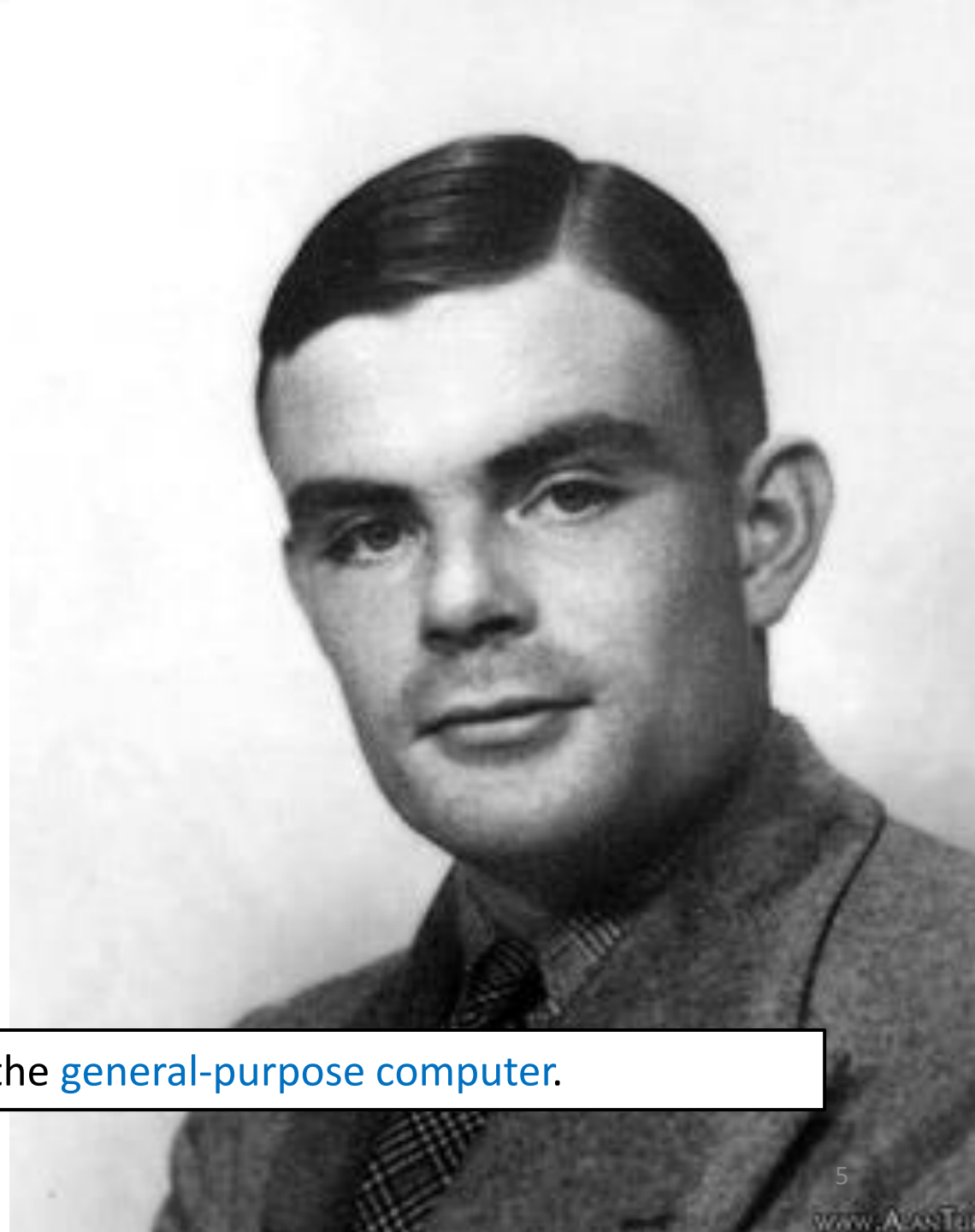
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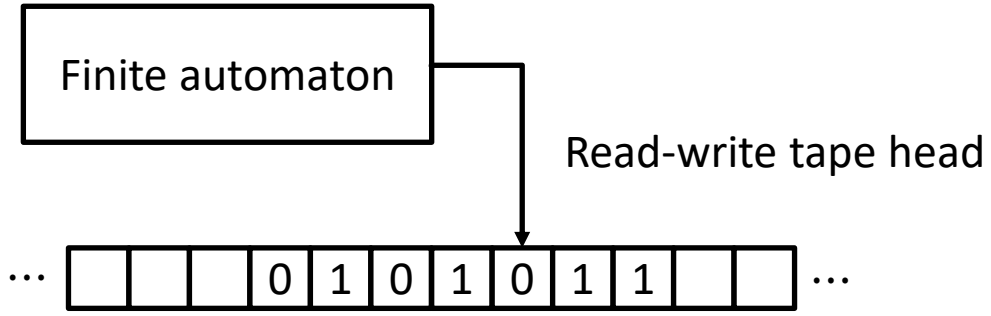
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A Turing machine is a theoretical model of the general-purpose computer.

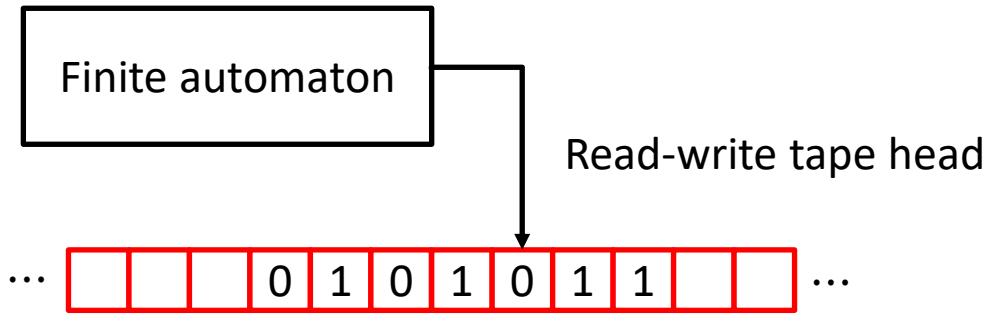
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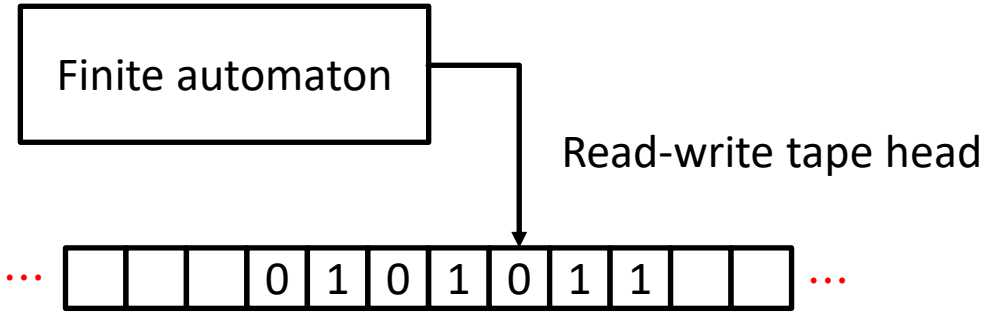




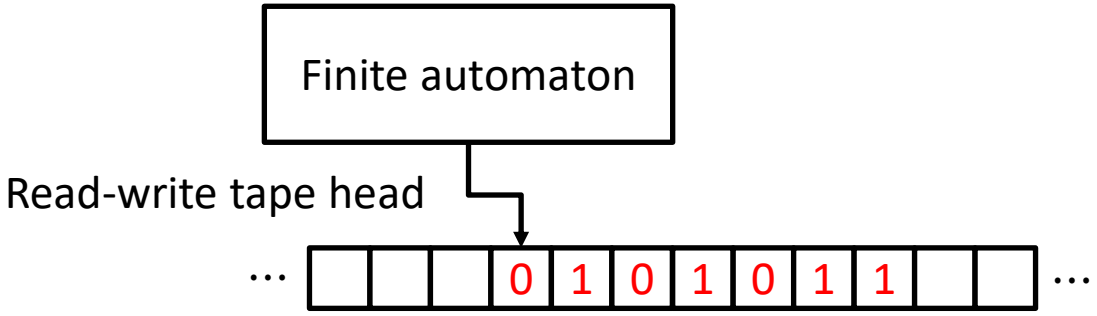
A Turing Machine is a (non-)deterministic finite automaton augmented with an infinite tape where symbols can be written and read.



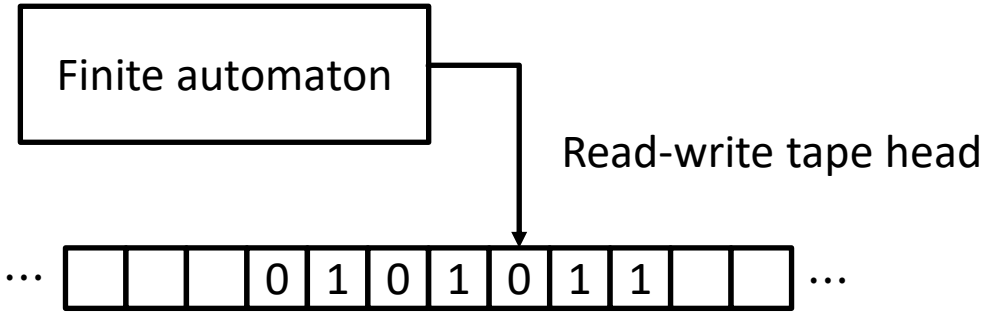
The tape serves as additional memory for the non-deterministic finite automaton (like the stack in a PDA).



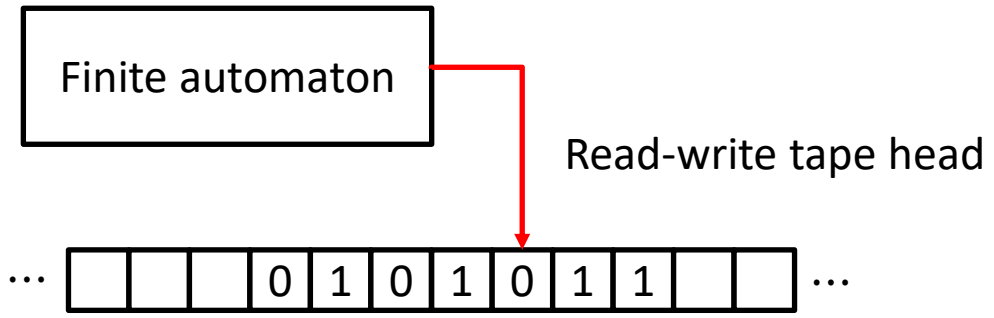
The tape is infinite in both directions.



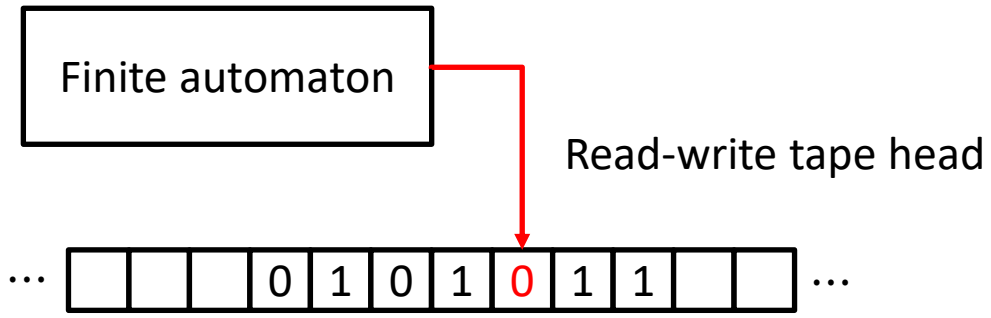
Initially, the input string is written on the tape with the tape head scanning the first symbol of the input.



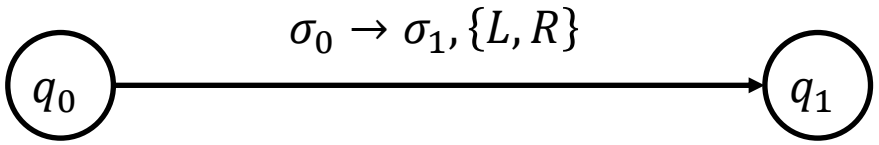
All other cells are filled with blanks, ␣, a special symbol that is not in the input alphabet.



The tape head can move left and right.



The tape head can read the symbol in the current cell and can overwrite the symbol with a new symbol.



We label transitions in Turing machines with a function (like PDAs).

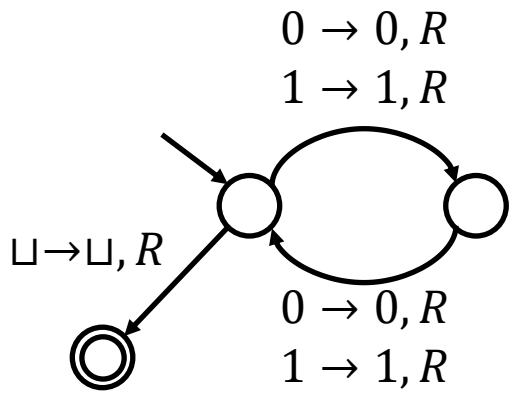
If σ_0 is scanned, then σ_1 is written in its place and the head moves left or right.

In Turing machines, the accept state and the reject state are two special states that once entered, ends computation.

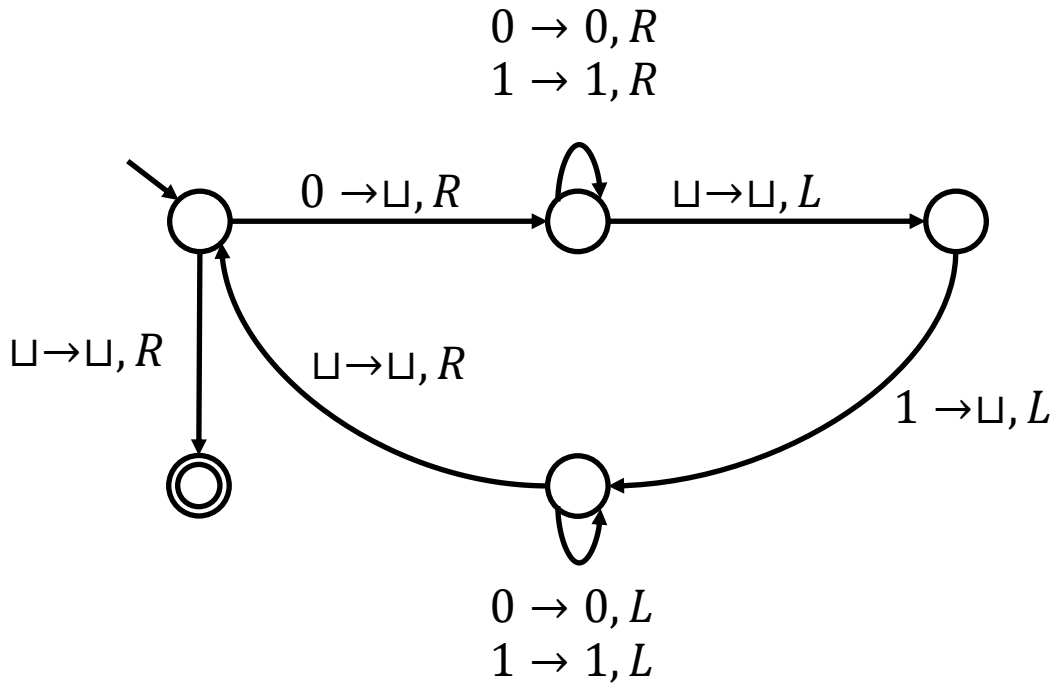
We typically don’t draw the reject state, as we assume all missing transitions lead there.

It is possible for a Turing machine to run forever, never reaching an accept or reject state. In this case we say it does not halt.

$$L = \{w \mid \text{the length of } w \text{ is even}\}$$



$$L = \{0^n 1^n \mid n \geq 0\}$$



Exercises

$L_0 = \{0^n 1^n 0^n \mid n \geq 0\}$

$L_1 = \{0^{2^n} \mid n \geq 0\}$

$L_2 = \{w\#w \mid w \in \{0,1\}^*\}$

The Theory of Computing

Formal Definition of Turing Machines

Definition: A deterministic Turing machine is a 7-tuple, $(Q, \Sigma, \Gamma, \delta, q_0, q_a, q_r)$, where Q, Σ, and Γ are finite sets and:

1. Q is the set of states,
2. Σ is the input alphabet not containing the blank symbol $\sqcup$,
3. Γ is the tape alphabet, where $\sqcup \in \Gamma$ and $\Sigma \subset \Gamma$,
4. $\delta: Q \times \Gamma \to Q \times \Gamma \times \{L, R\}$ is the transition function,
5. $q_0 \in Q$ is the start state,
6. $q_a \in Q$ is the accept state, and
7. $q_r \in Q$ is the reject state where $q_r \neq q_a$.

Definition: A configuration of a Turing machine is a combination of the current state, the current tape contents and the location of the tape head.

By convention, we often write a configuration by writing the non-blank symbols of the tape with the state before the symbol scanned by the tape head. For example:

$$1010q_7 0011$$

Definition: The start configuration of M is state q_0 with the input string

$$w = w_1 w_2 \cdots w_n \in \Sigma^*$$

written on the tape, and with the tape head scanning symbol w_1. All other cells contain $\sqcup \in \Gamma$.

$$q_0 w_1 w_2 \cdots w_n$$

Definition: An accepting configuration is any configuration in state q_a.

Definition: A rejecting configuration is any configuration in state q_r.

Definition: A halting configuration is any accepting or rejecting configuration.

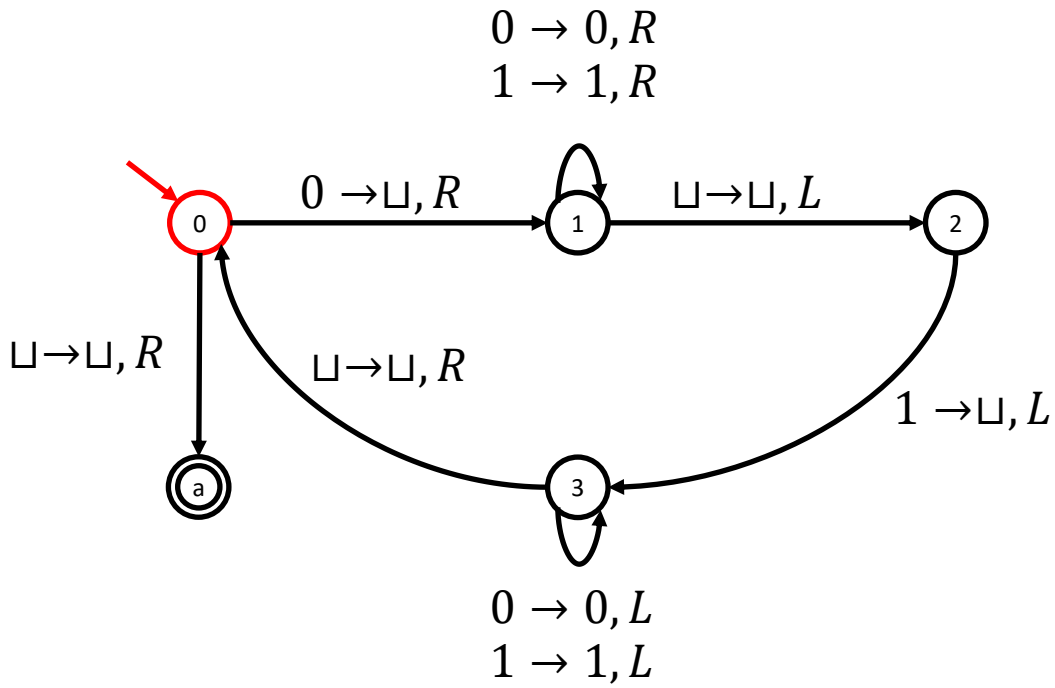
Definition: A configuration C_1 yields C_2 if M can move from C_1 to C_2 in a single step.

Definition: A Turing machine M accepts input w if a sequence of configurations $C_1, C_2, \ldots C_k$ exists, where C_1 is the start configuration of M on w, each C_i yields C_{i+1}, and C_k is an accepting configuration. We call $C_1, C_2, \ldots C_k$ an accepting computation of M on w.

Definition: A Turing machine M rejects input w if a sequence of configurations $C_1, C_2, \ldots C_k$ exists, where C_1 is the start configuration of M on w, each C_i yields C_{i+1}, and C_k is a rejecting configuration. We call $C_1, C_2, \ldots C_k$ a rejecting computation of M on w.

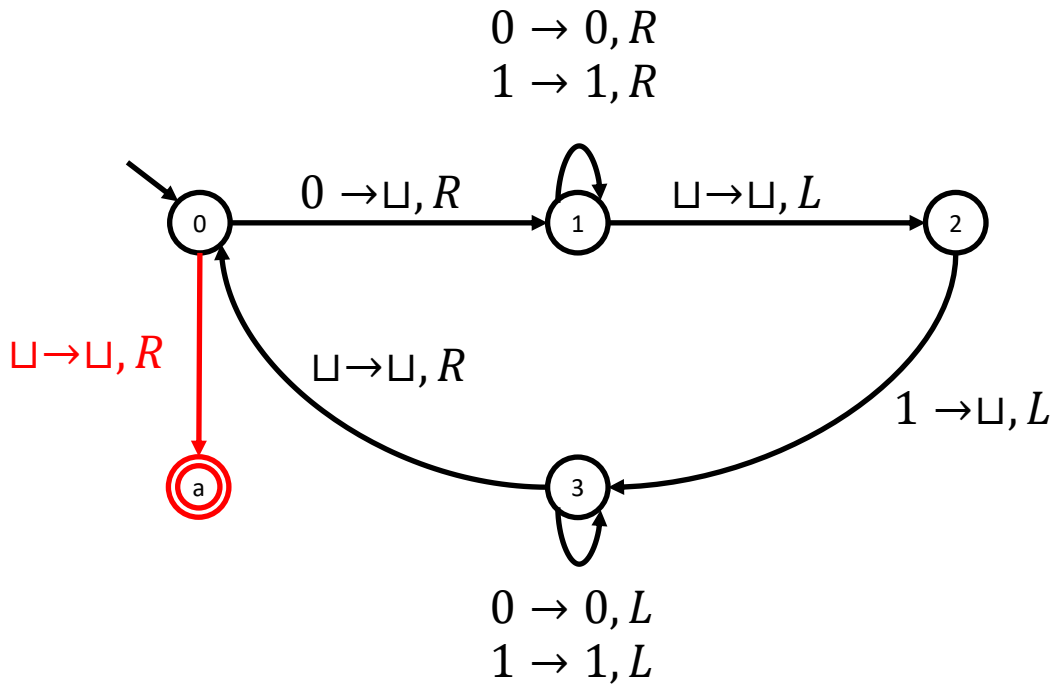
We can trace the computation of this machine on the string $w = \epsilon$.

q_0



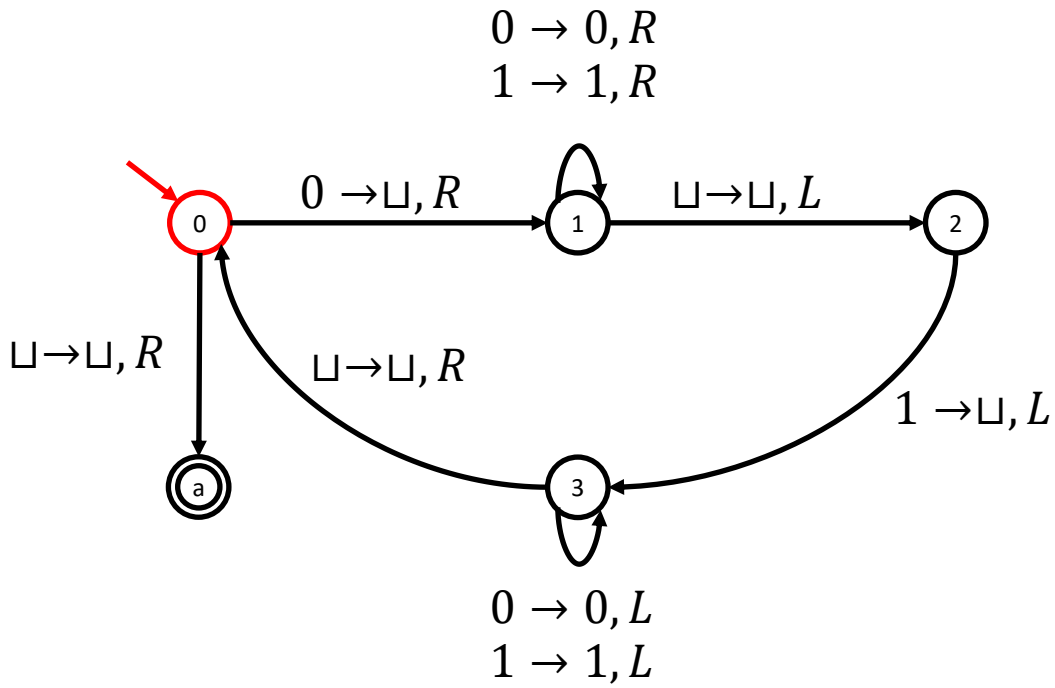
We can trace the computation of this machine on the string $w = \epsilon$.

$q_0 \Rightarrow q_a$



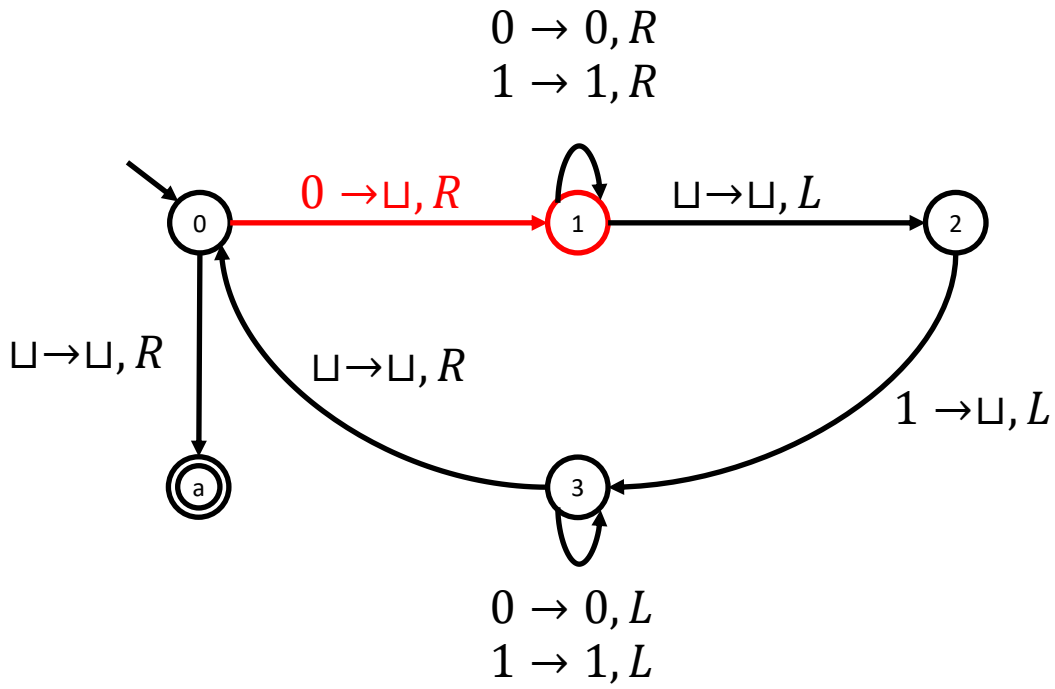
We can trace the computation of this machine on the string $w = 001$.

$q_0 001$



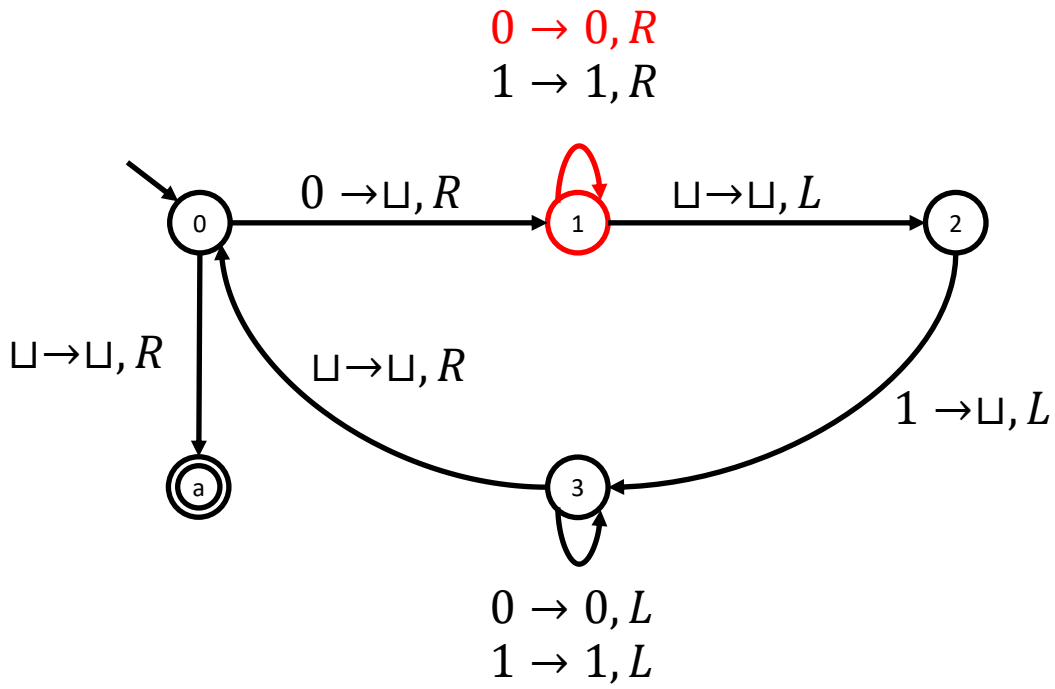
We can trace the computation of this machine on the string $w = 001$.

$q_0001 \Rightarrow q_101$



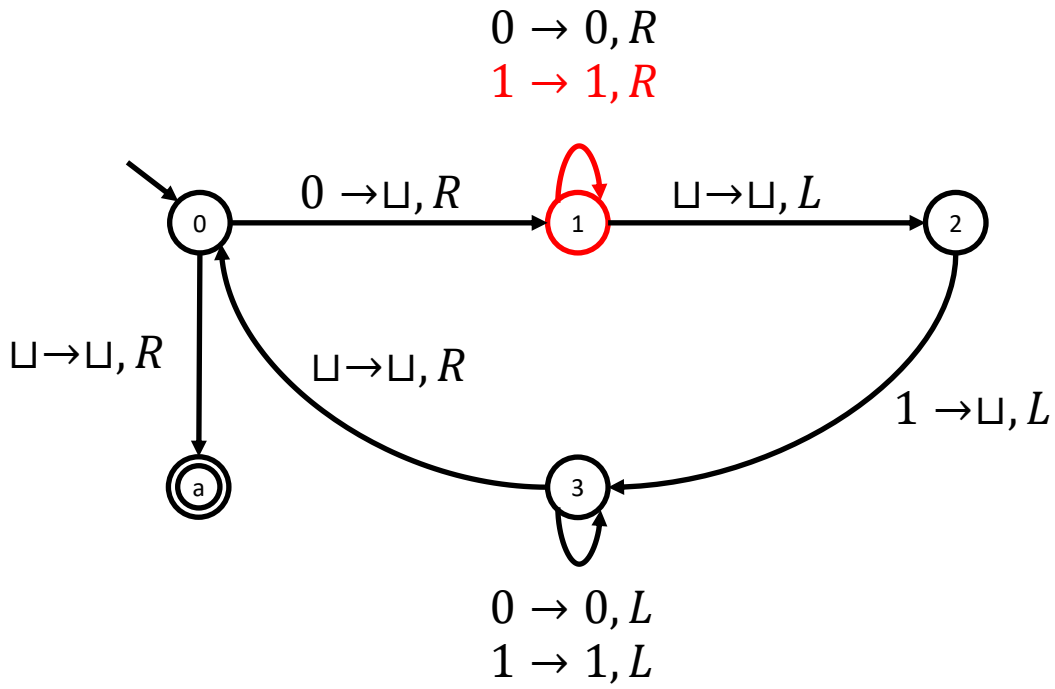
We can trace the computation of this machine on the string $w = 001$.

$$q_0001 \Rightarrow q_101 \Rightarrow 0q_11$$



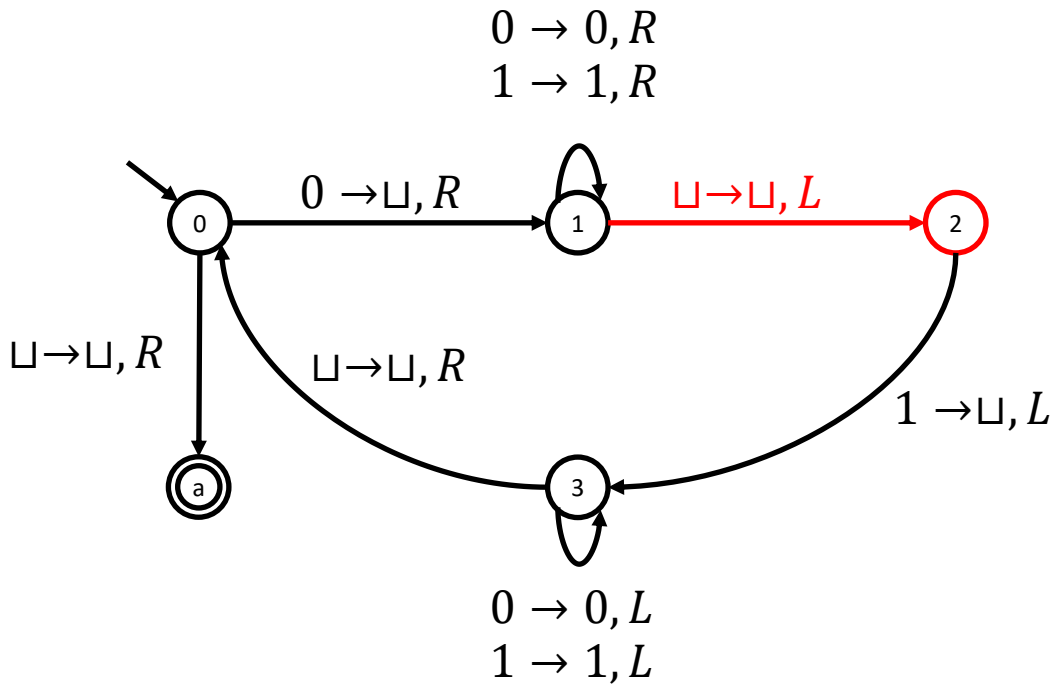
We can trace the computation of this machine on the string $w = 001$.

$$q_0001 \Rightarrow q_101 \Rightarrow 0q_11 \Rightarrow 01q_1$$



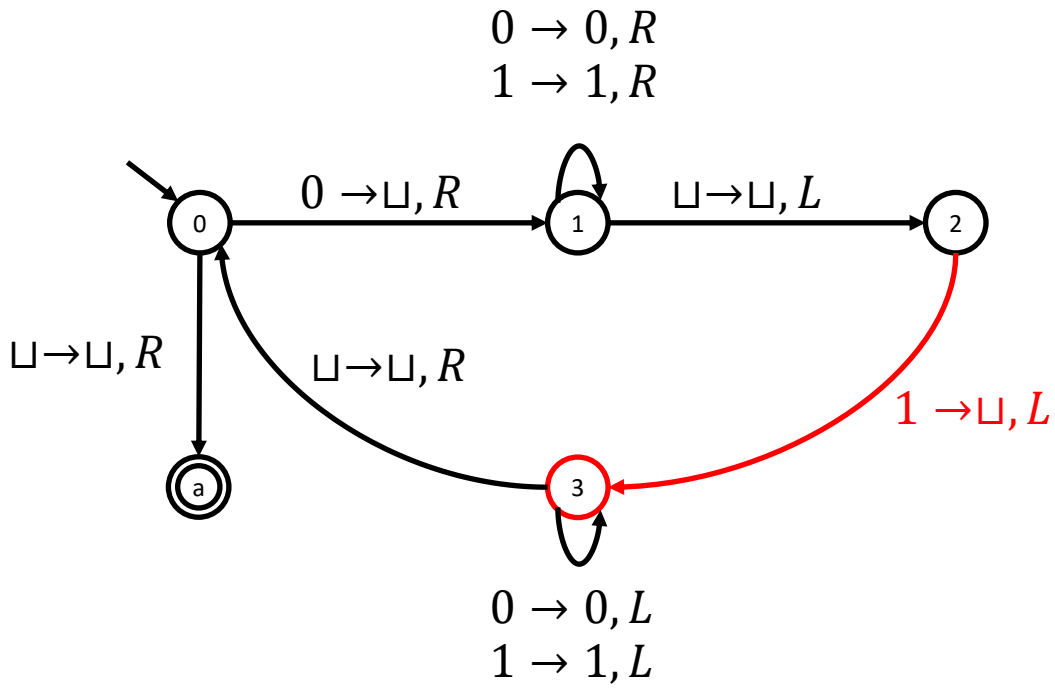
We can trace the computation of this machine on the string $w = 001$.

$$q_0001 \Rightarrow q_101 \Rightarrow 0q_11 \Rightarrow 01q_1 \Rightarrow 0q_21$$



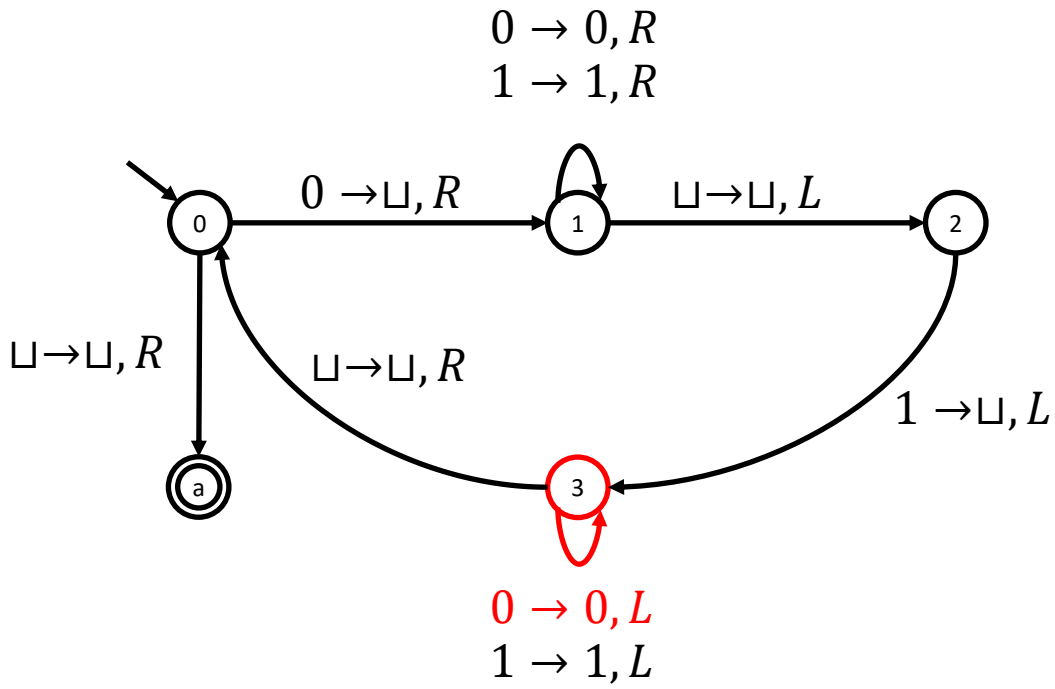
We can trace the computation of this machine on the string $w = 001$.

$$q_0001 \Rightarrow q_101 \Rightarrow 0q_11 \Rightarrow 01q_1 \Rightarrow 0q_21 \Rightarrow q_30$$



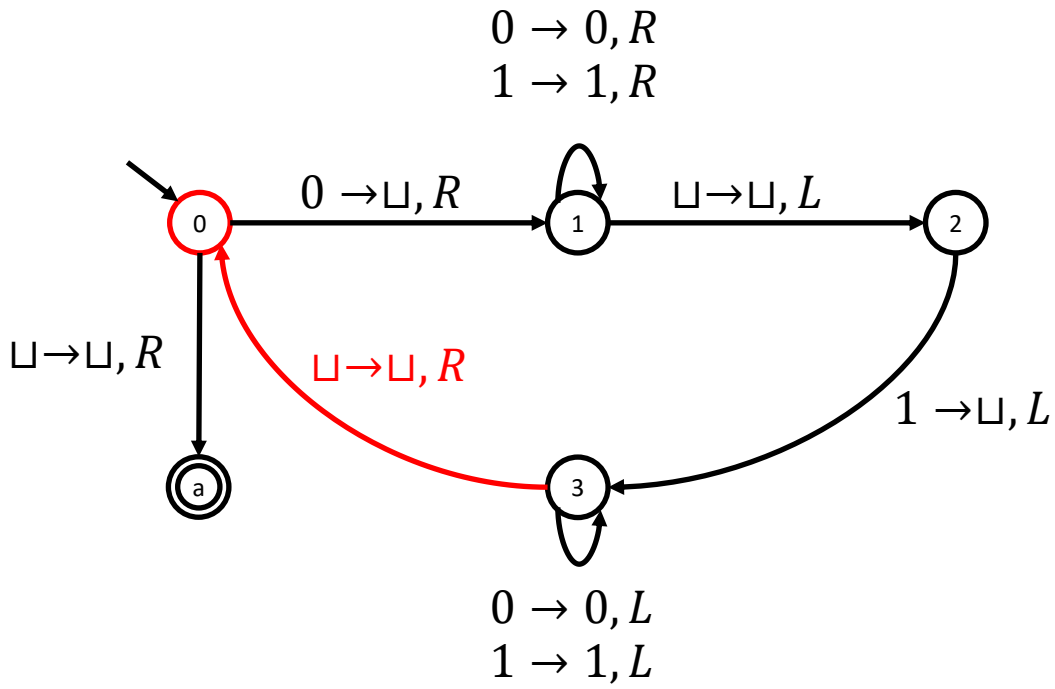
We can trace the computation of this machine on the string $w = 001$.

$$q_0001 \Rightarrow q_101 \Rightarrow 0q_11 \Rightarrow 01q_1 \Rightarrow 0q_21$$
$$\Rightarrow q_30 \Rightarrow q_3_0$$



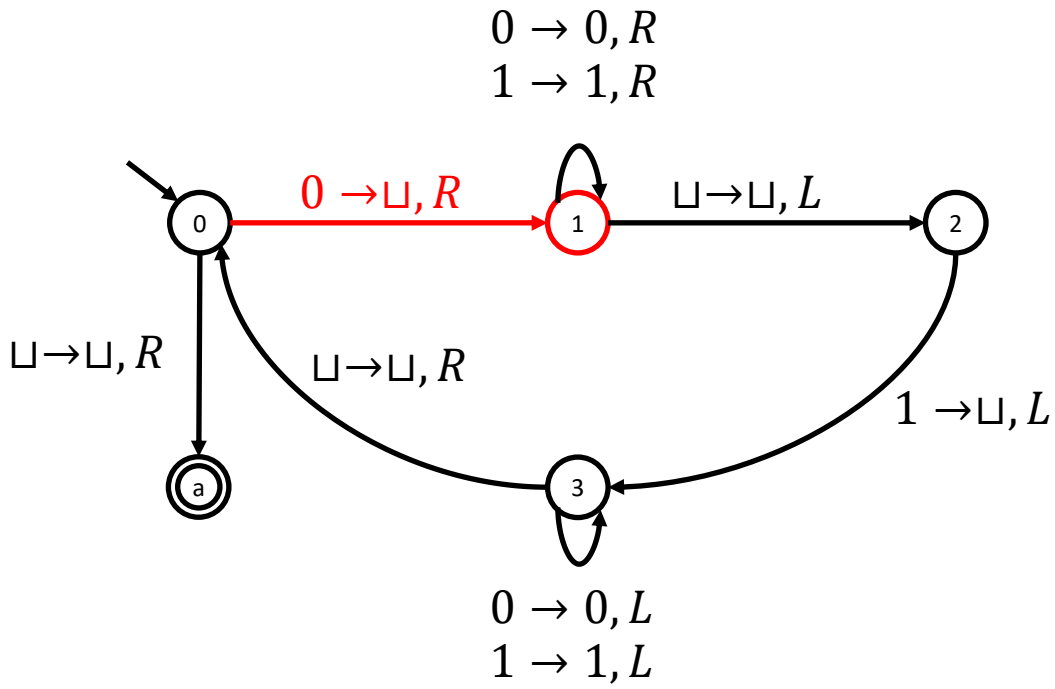
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$$\Rightarrow q_30 \Rightarrow q_3_0 \Rightarrow q_00$$



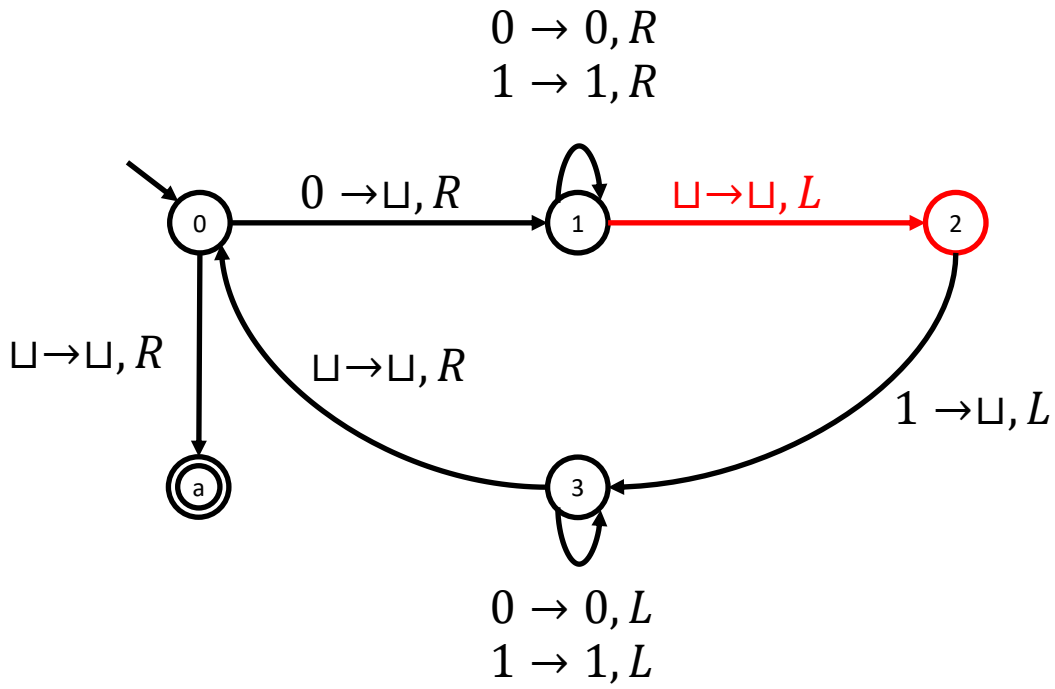
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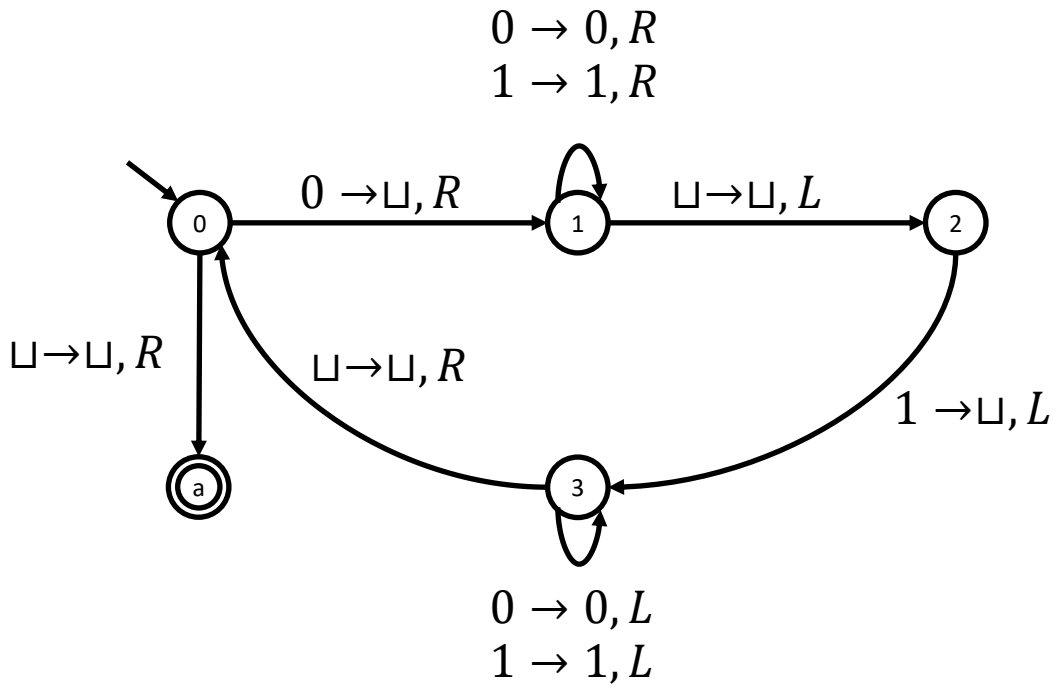
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$$q_0001 \Rightarrow q_101 \Rightarrow 0q_11 \Rightarrow 01q_1 \Rightarrow 0q_21$$
$$\Rightarrow q_30 \Rightarrow q_3_0 \Rightarrow q_00 \Rightarrow q_1 \Rightarrow q_2$$



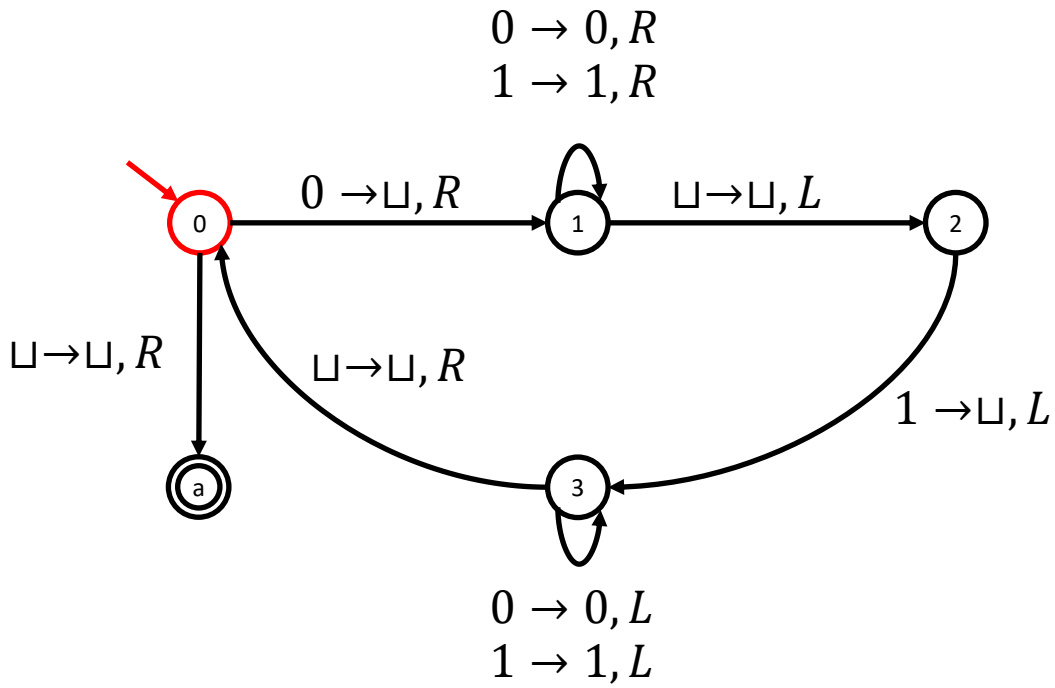
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$$\Rightarrow q_30 \Rightarrow q_3_0 \Rightarrow q_00 \Rightarrow q_1 \Rightarrow q_2 \Rightarrow q_r$$



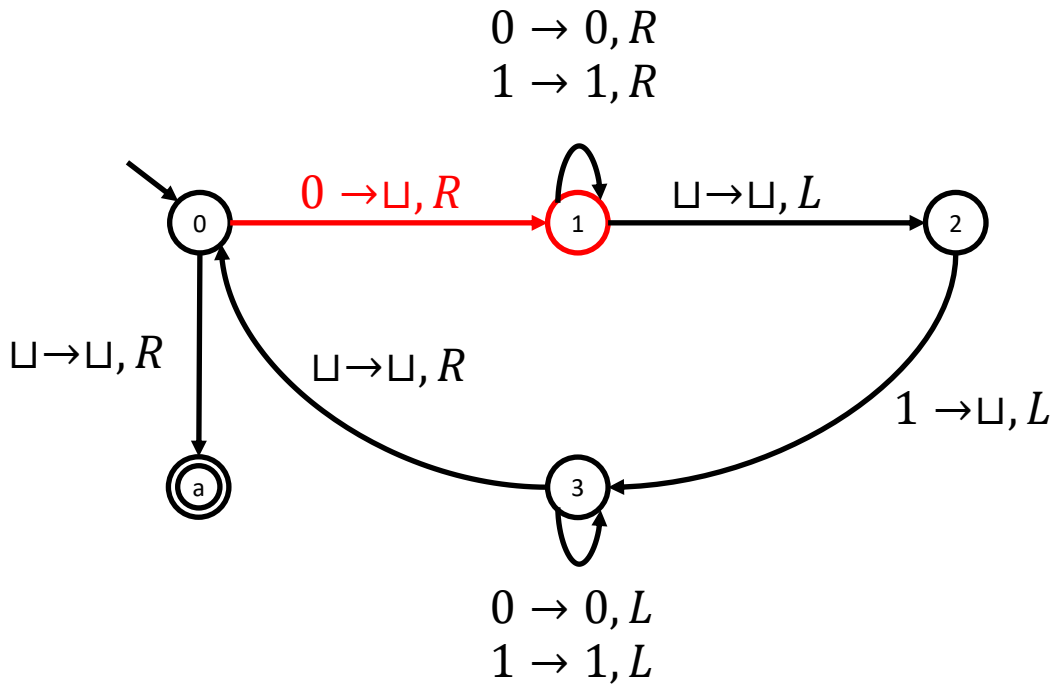
We can trace the computation of this machine on the string $w = 0011$.

$q_0 0011$



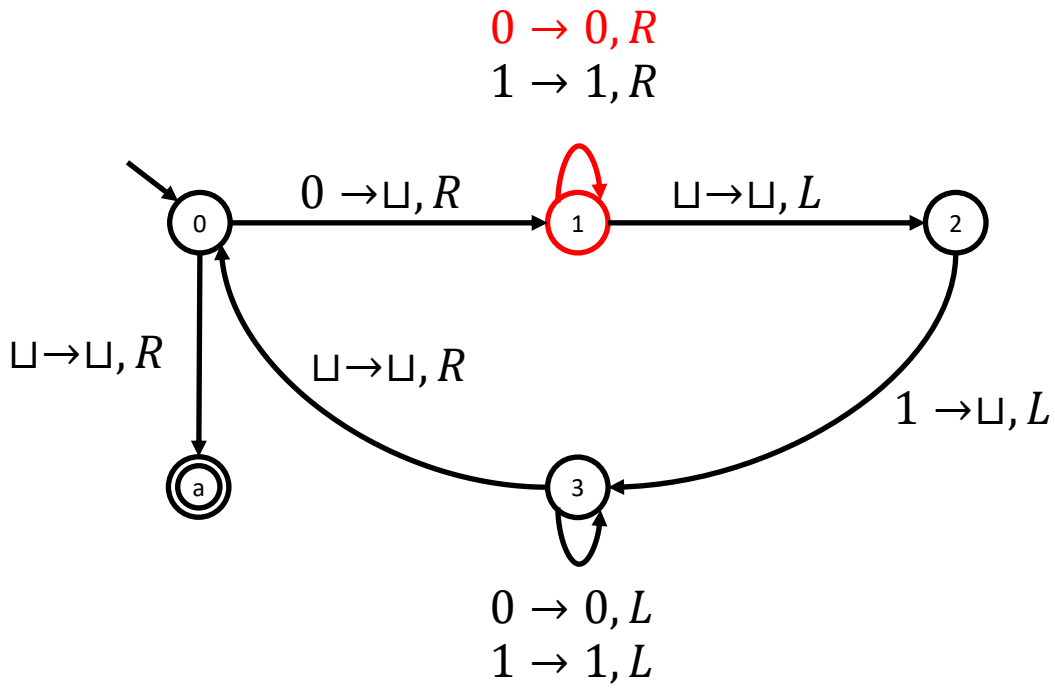
We can trace the computation of this machine on the string $w = 0011$.

$q_0 0011 \Rightarrow q_1 011$



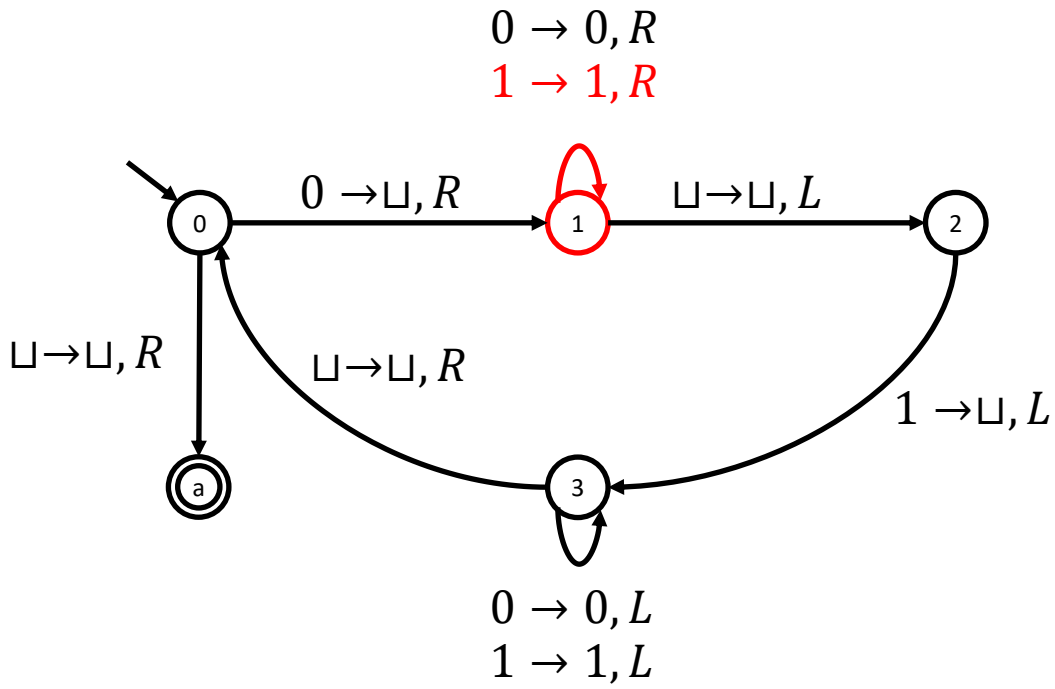
We can trace the computation of this machine on the string $w = 0011$.

$$q_0 0011 \Rightarrow q_1 011 \Rightarrow 0 q_1 11$$



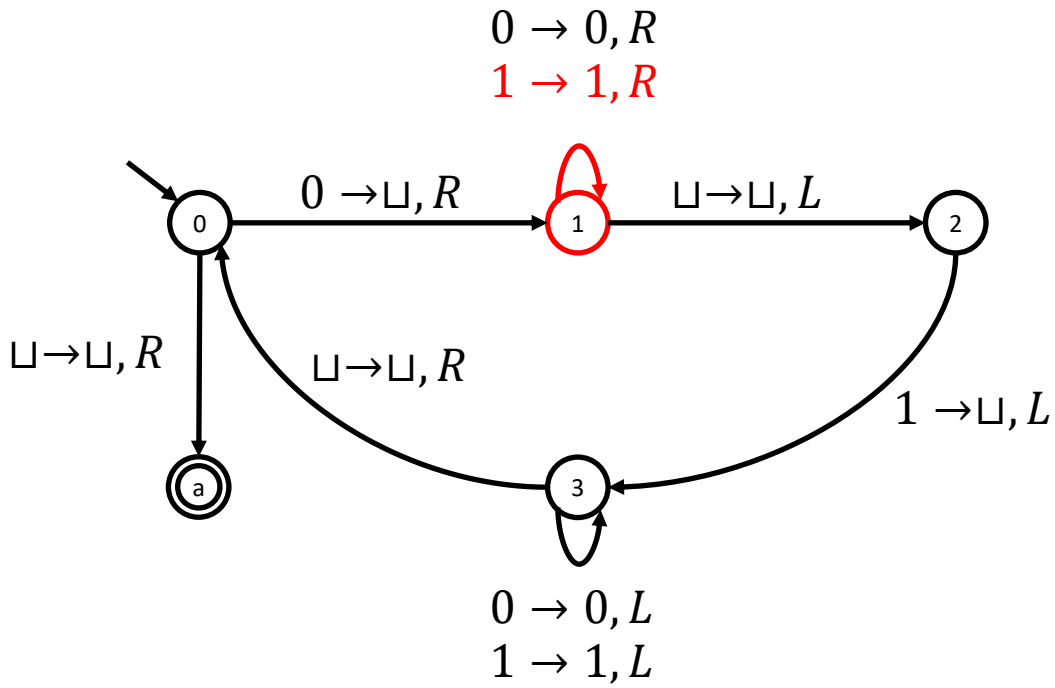
We can trace the computation of this machine on the string $w = 0011$.

$$q_0 0011 \Rightarrow q_1 011 \Rightarrow 0q_1 11 \Rightarrow 01q_1 1$$



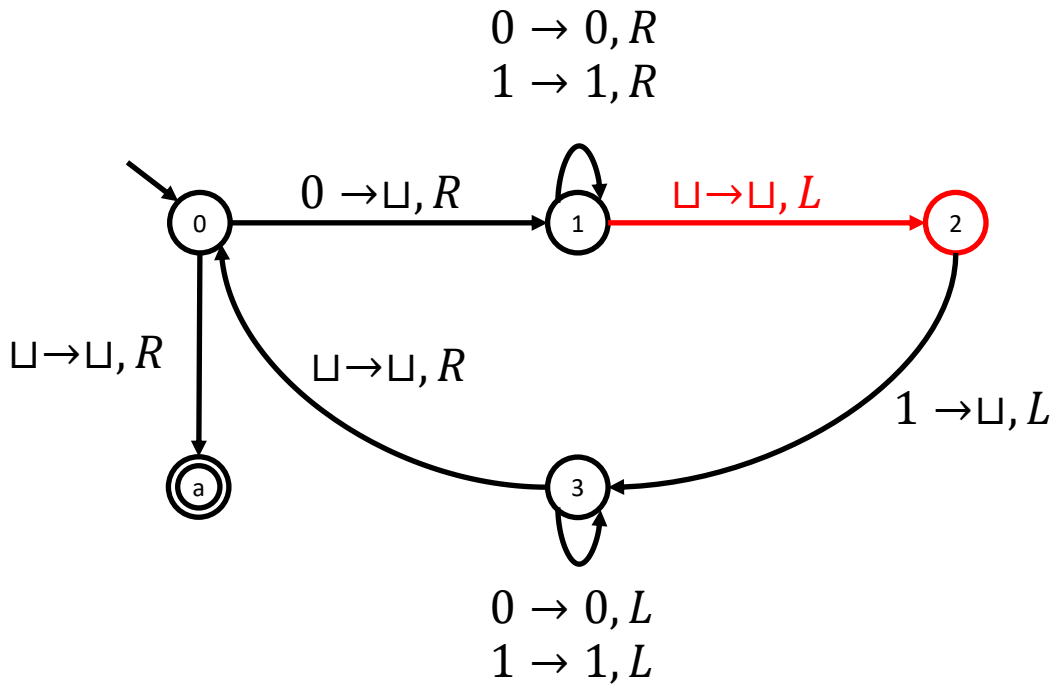
We can trace the computation of this machine on the string $w = 0011$.

$$q_0 0011 \Rightarrow q_1 011 \Rightarrow 0q_1 11 \Rightarrow 01q_1 1$$
$$\Rightarrow 011q_1$$



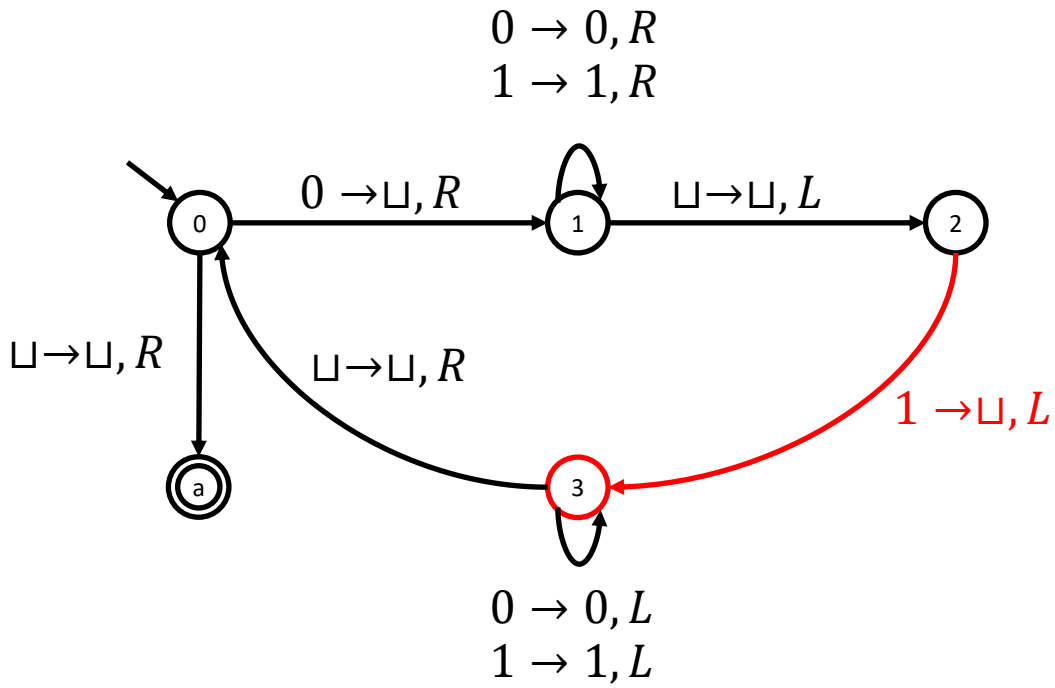
We can trace the computation of this machine on the string $w = 0011$.

$$q_0 0011 \Rightarrow q_1 011 \Rightarrow 0q_1 11 \Rightarrow 01q_1 1$$
$$\Rightarrow 011q_1 \Rightarrow 01q_2 1$$



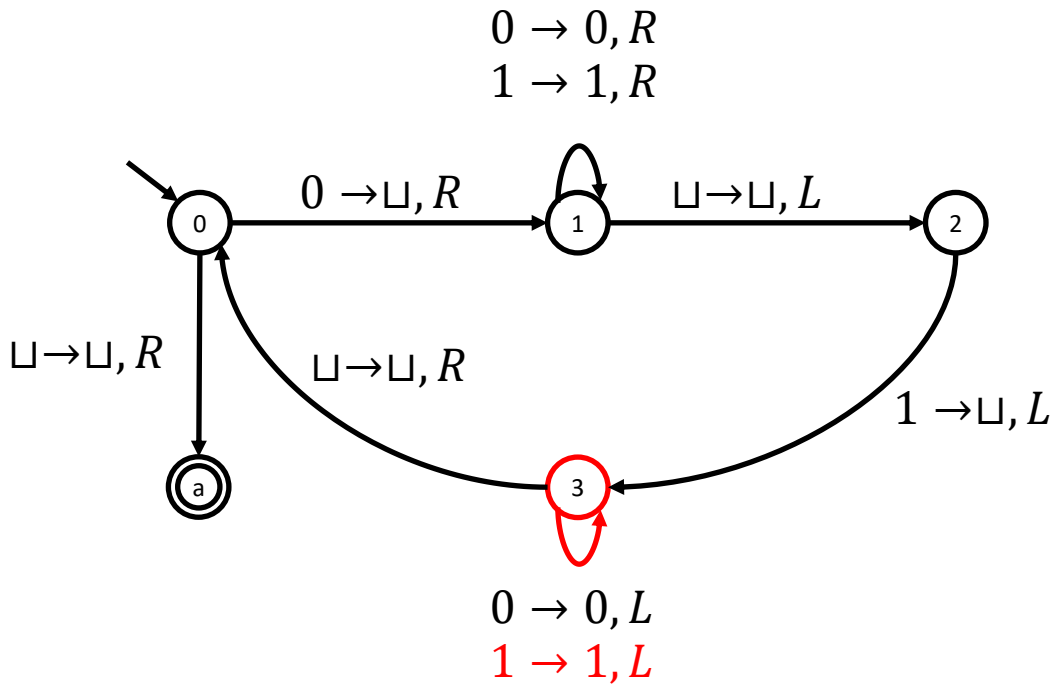
We can trace the computation of this machine on the string $w = 0011$.

$$q_0 0011 \Rightarrow q_1 011 \Rightarrow 0q_1 11 \Rightarrow 01q_1 1$$
$$\Rightarrow 011q_1 \Rightarrow 01q_2 1 \Rightarrow 0q_3 1$$



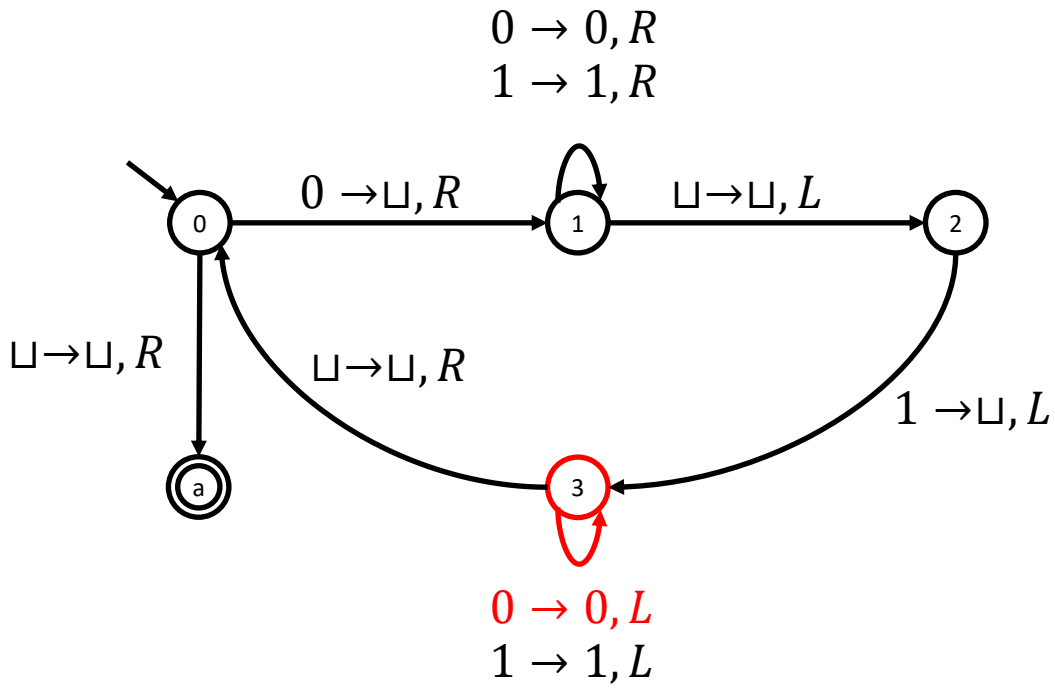
We can trace the computation of this machine on the string $w = 0011$.

$$q_0 0011 \Rightarrow q_1 011 \Rightarrow 0q_1 11 \Rightarrow 01q_1 1$$
$$\Rightarrow 011q_1 \Rightarrow 01q_2 1 \Rightarrow 0q_3 1 \Rightarrow q_3 01$$



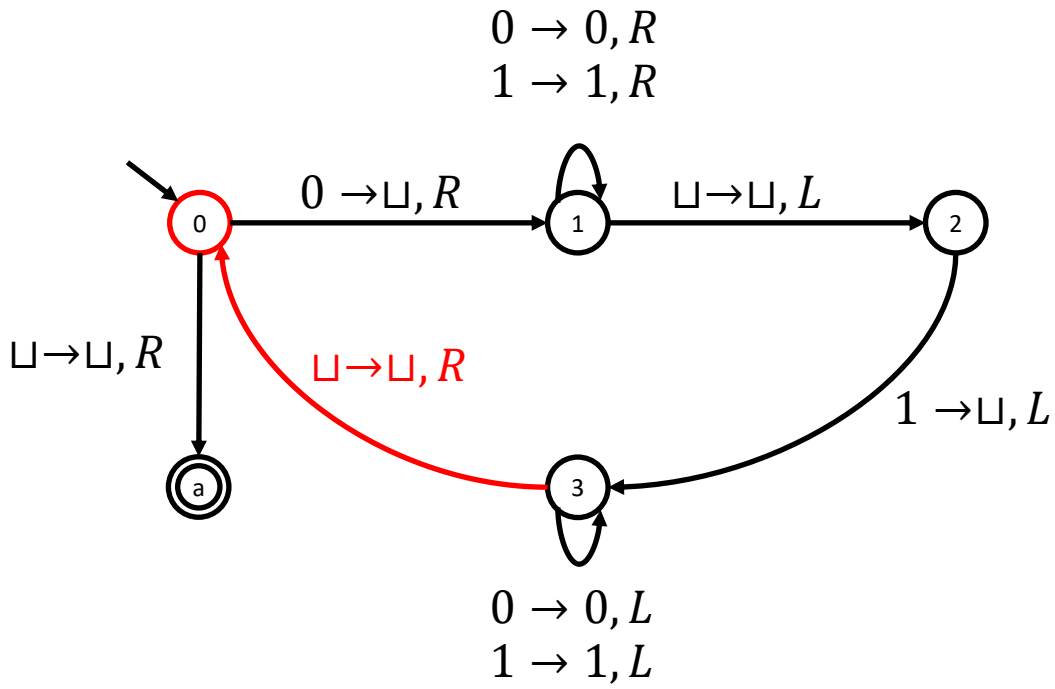
We can trace the computation of this machine on the string $w = 0011$.

$$q_0 0011 \Rightarrow q_1 011 \Rightarrow 0q_1 11 \Rightarrow 01q_1 1$$
$$\Rightarrow 011q_1 \Rightarrow 01q_2 1 \Rightarrow 0q_3 1 \Rightarrow q_3 01$$
$$\Rightarrow q_3 _01$$



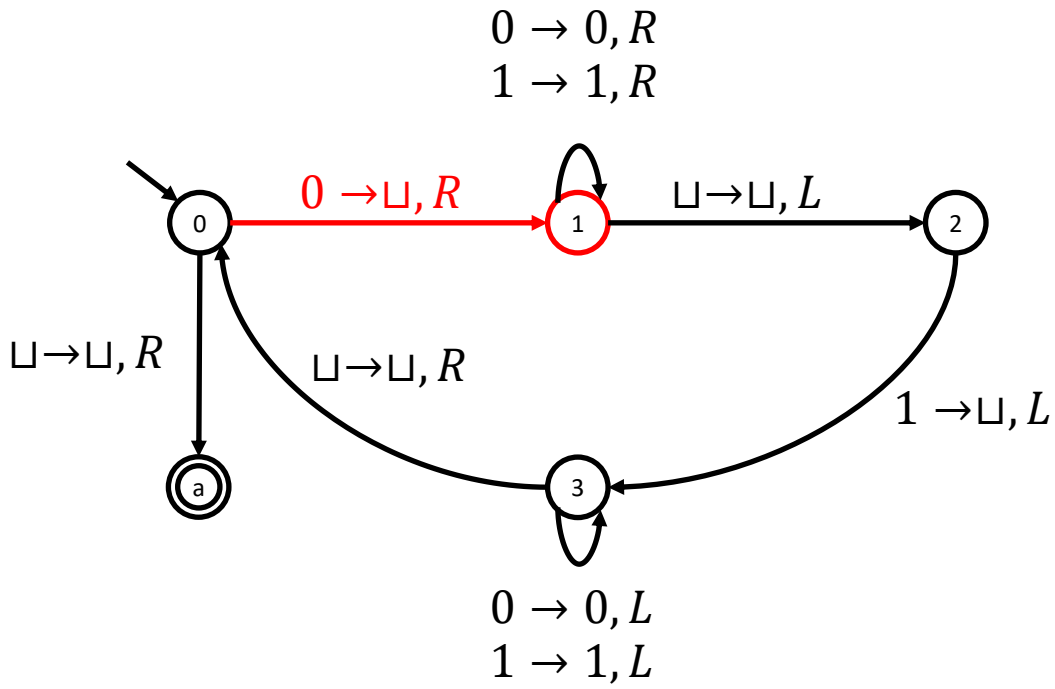
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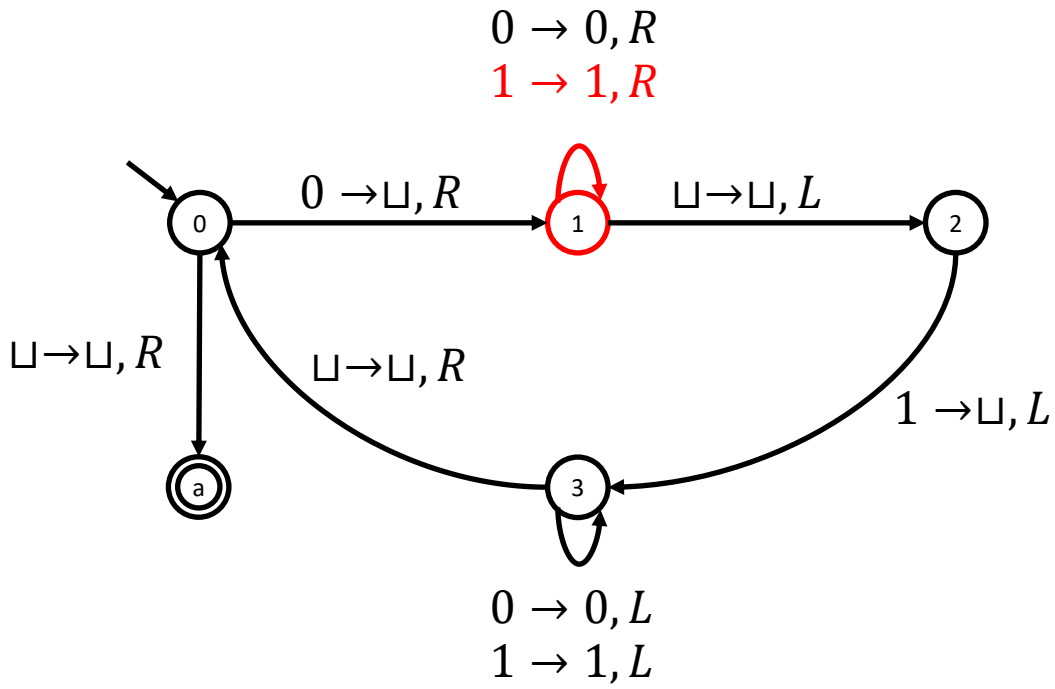
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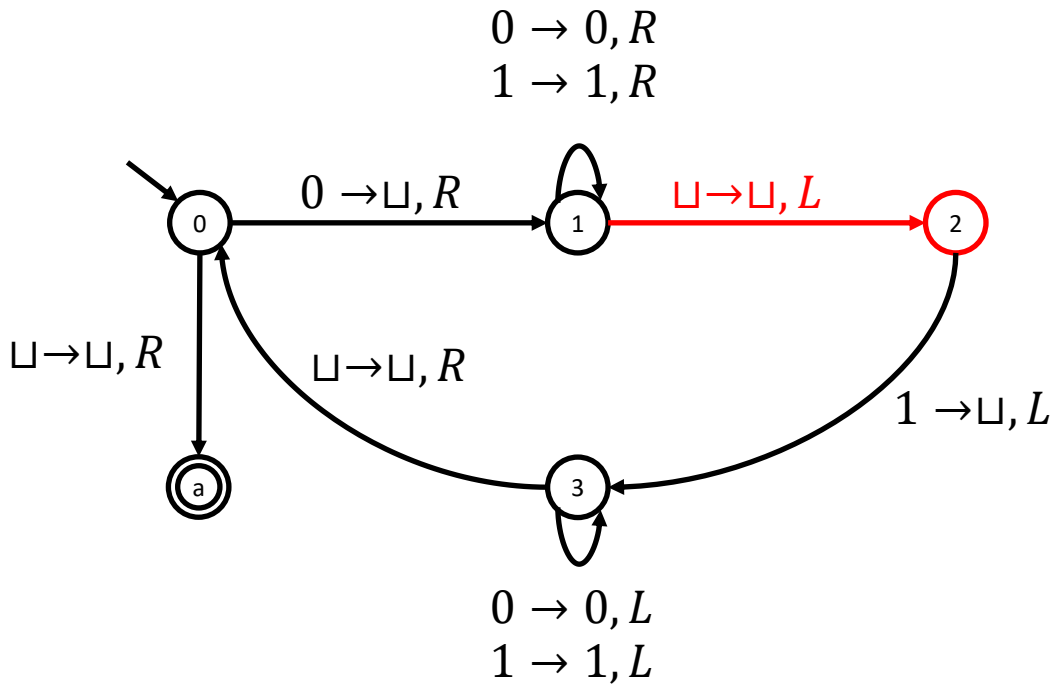
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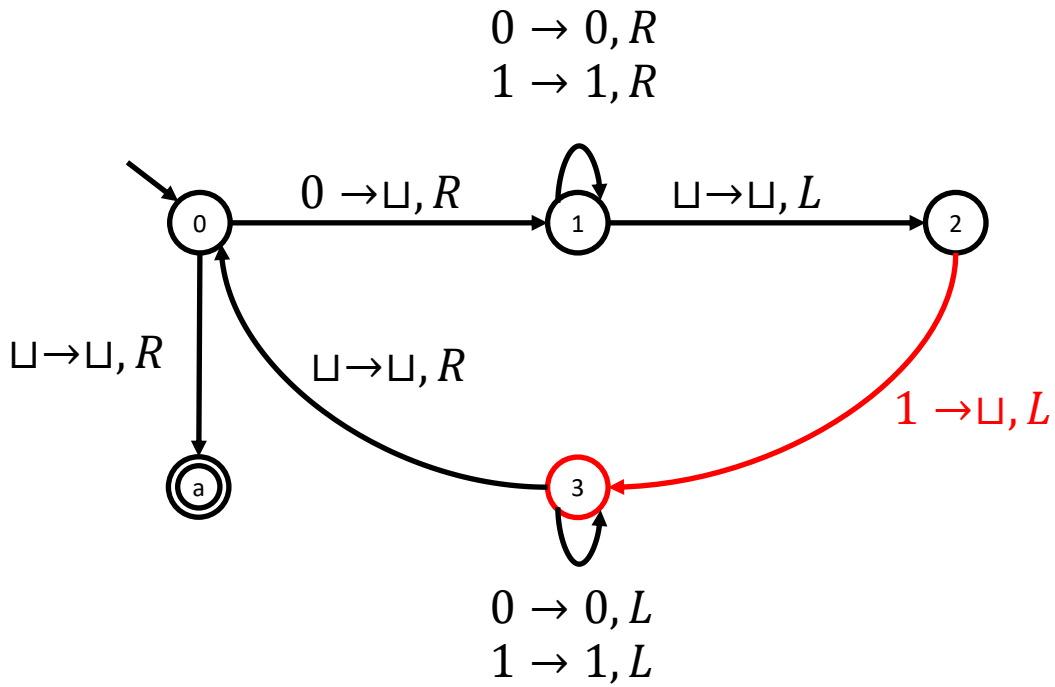
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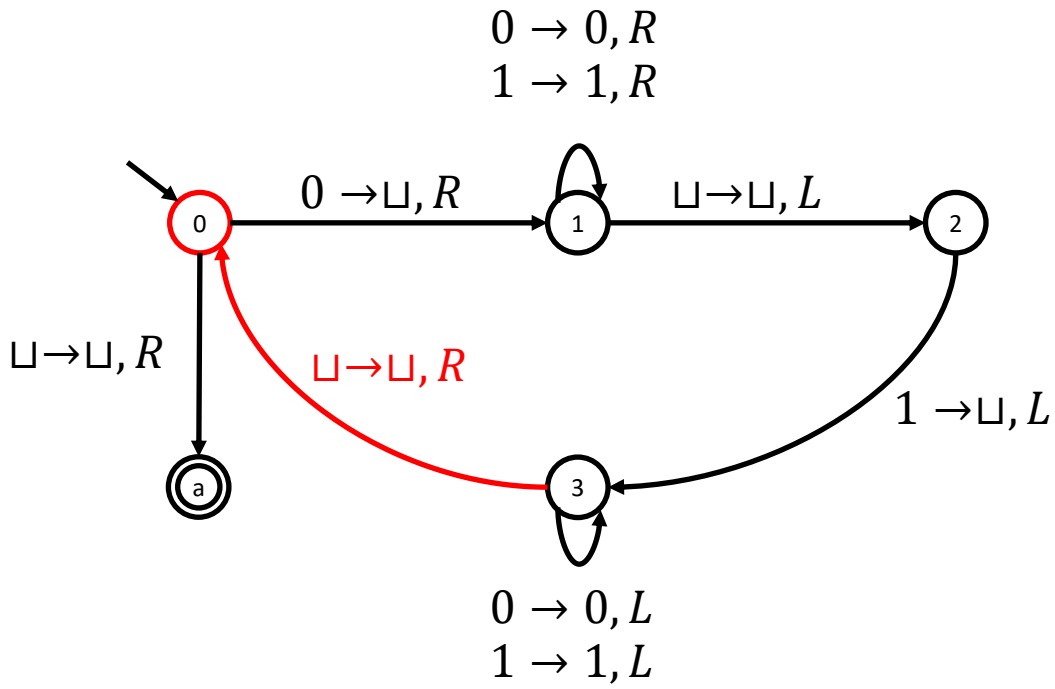
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$$\Rightarrow q_3$$



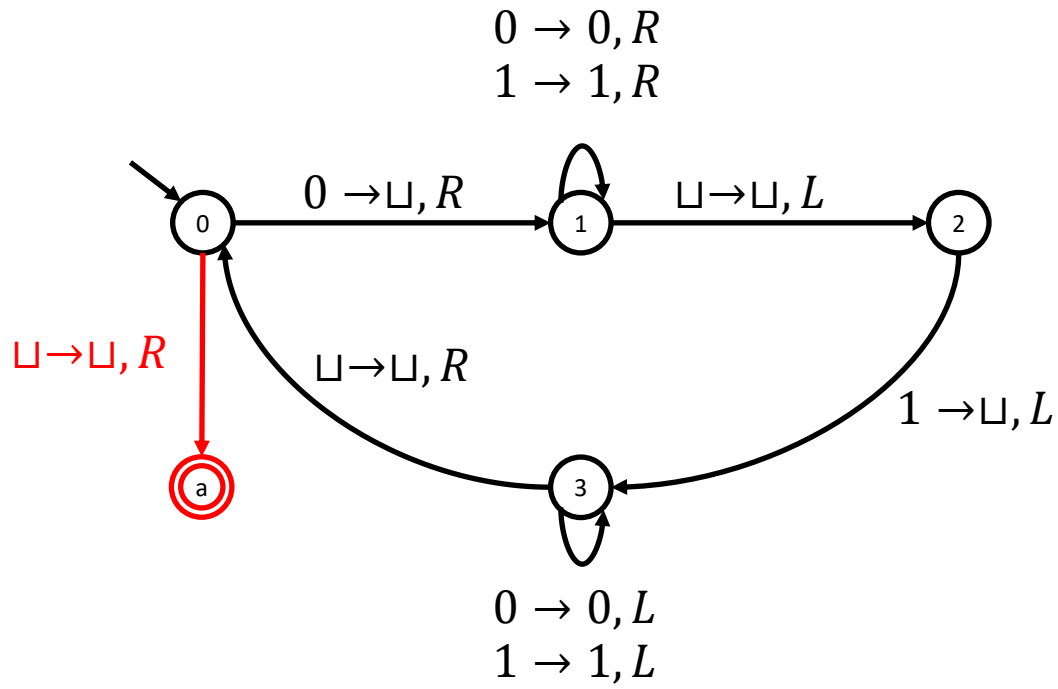
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$$\Rightarrow q_3 _01 \Rightarrow q_0 01 \Rightarrow q_1 1 \Rightarrow 1q_1 \Rightarrow q_2 1$$
$$\Rightarrow q_3 \Rightarrow q_0$$



We can trace the computation of this machine on the string $w = 001$.

$$q_0 0011 \Rightarrow q_1 011 \Rightarrow 0q_1 11 \Rightarrow 01q_1 1$$
$$\Rightarrow 011q_1 \Rightarrow 01q_2 1 \Rightarrow 0q_3 1 \Rightarrow q_3 01$$
$$\Rightarrow q_3 _01 \Rightarrow q_0 01 \Rightarrow q_1 1 \Rightarrow 1q_1 \Rightarrow q_2 1$$
$$\Rightarrow q_3 \Rightarrow q_0 \Rightarrow q_a$$



Definition: $L(M)$, the language of M, also called the language recognized by M, is

$$\{w \in \Sigma^* \mid w \text{ is accepted by } M\}.$$

Definition: A language L is Turing recognizable if there exists a Turing machine M that recognizes it.

Notice that just because a Turing machines accepts a set of strings it does not mean that it rejects the other strings.

Definition: A Turing machine M decides L if M accepts all $w \in L$ and rejects all $w \notin L$.

Definition: A language L is Turing decidable if there exists a Turing machine M that decides it.

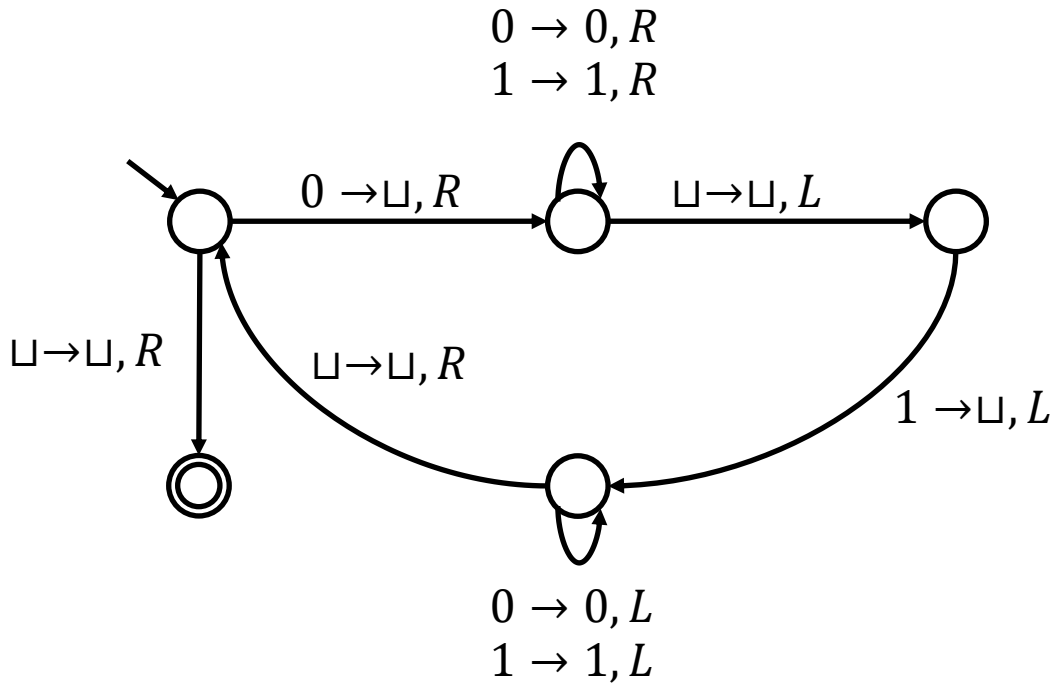
The Theory of Computing

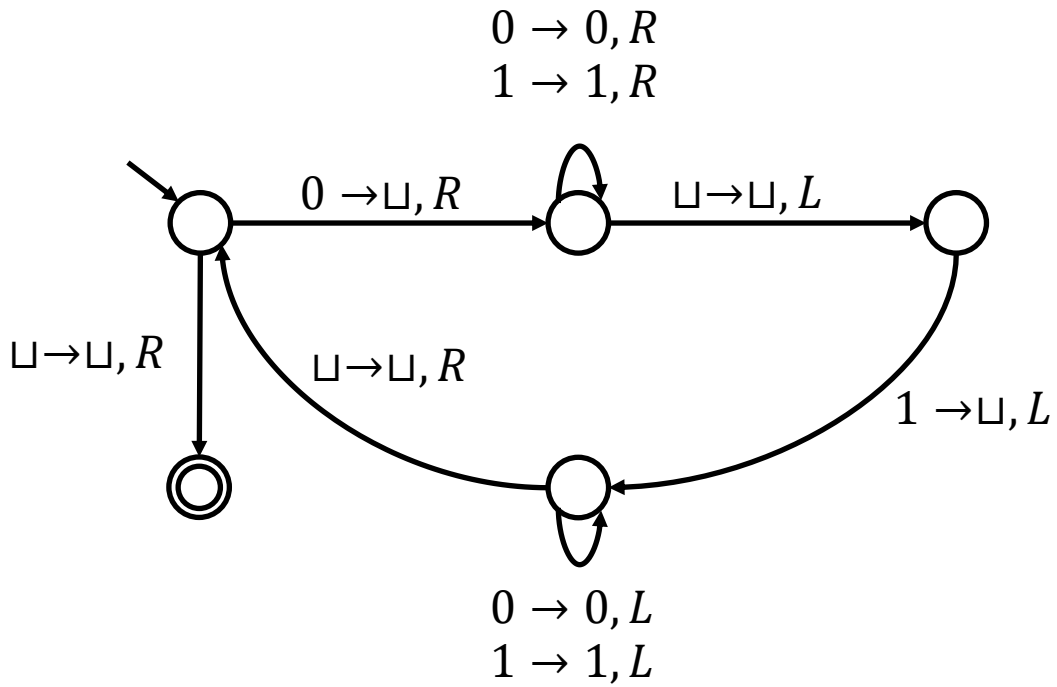
Pseudocode for Turing Machine

Turing machine state diagrams can be very large and drawing them can be tedious.

Except for the simplest Turing machines, we use a pseudocode to describe machines instead of formal diagrams.

$$L = \{0^n 1^n \mid n \geq 0\}$$





M_1 = "On input w:

1. If the symbol scanned is a blank, accept. If the symbol scanned is 1, reject.
2. Replace the first 0 with a ⊔ and then reposition to the last non-blank symbol. If none exists, reject. If the symbol is a 0, reject.
3. Replace the last 1 with a ⊔ and then reposition to the first non-blank symbol. If none exists, accept. If the symbol is a 1, reject.
4. Go to step 2."

$$L_2 = \{w\#w \mid w \in \{0,1\}^*\}$$

M_2 = "On input $w\#x$ where $w, x \in \{0,1\}^*$:

1. If w and x contain only Xs, accept. If either x or w contains only Xs and the other contains 0s or 1s, reject.
2. Replace the first 0 or 1 of w with an X and remember it.
3. Then scan to the first 0 or 1 after the #. If it matches the remembered symbol, replace it with an X, otherwise reject.
4. Reposition to the first symbol of w.
5. Go to step 1."

Turing machines are capable of computing mathematical functions.

$$L_+ = \{0^n \# 0^m \# 0^{n+m} \mid n, m \geq 0\}$$

$$L_- = \{0^n \# 0^m \# 0^{n-m} \mid n \geq m \geq 0\}$$

These languages are context free, and so can be recognized using a push-down automata.

(Left as an exercise.)

$$L_{\times} = \{0^n \# 0^m \# 0^{n \cdot m} \mid n, m \geq 0\}$$

$M_{\times}$ = "On input $0^* \# 0^* \# 0^*$:

1. Replace a single 0 in the second block with an X. If none remain and the last block contains no 0s, accept. If none remain and the last block contains 0s, reject.
2. Moving between the first and last block, replace n 0s with Xs in both blocks. If the last block runs out of 0s in this step, reject. Once done, replace the Xs in the first block with 0s for the next loop.
3. Go to step 1."

$$L_{\square} = \left\{ 0^n \# 0^{n^2} \mid n \geq 0 \right\}$$

$M_{\square}$ = "On input $0^* \# 0^*$:

1. Place a # before the input and then copy the first block before the # by replacing each 0 in the block with Xs. Restore the Xs to 0s when done and reposition to the start of the new input. After this step the tape should contain $0^n \# 0^n \# 0^m$.
2. Run $M_{\times}$ on the current input and accept or reject as it does."

$$L_{\square\square} = \left\{0^{n^2} \mid n \geq 0\right\}$$

$M_{\square\square}$ = "On input 0^*:

1. Place '$0\#$' before the input.
2. Run $M_{\square}$ on the current input and accept if it does. If it rejects and the first block of 0s is larger than the second, reject.
3. Add a single 0 to the first block and go to step 2."

With this technique well-defined integer functions can be recognized by Turning machines.

$$L_f = \{0^{f(n)} \mid n \geq 0\}$$